

SamPave Engineering Training Guide — V2.0

Professional Baseline

CRITICAL SAFETY DISCLAIMER

IMPORTANT:

SamPave is a professional decision-support engineering tool. All empirical design projections, thickness suggestions, material specifications, and mechanistic calculations generated by this software are for engineering screening and decision-support only. Final pavement designs, thickness profiles, and mix configurations must be independently verified, reviewed, and signed off by a licensed and qualified Professional Pavement Engineer before field execution, bidding, or construction.

1. Pavement Design Philosophy (IRC:37-2018)

IRC:37-2018 guides the design of flexible pavements using a mechanistic-empirical (M-E) framework. Pavements are modeled as multi-layered elastic structures under a standard dual-wheel configuration. Design evaluation involves two core criteria:

1. **Bituminous layer fatigue cracking** (tensile strain at the bottom of the bituminous layer, ϵ_t).
2. **Subgrade rutting** (vertical compressive strain at the top of the subgrade, ϵ_v).

2. Mathematical Core & Structural Equations

2.1 Traffic Projections (Cumulative MSA)

Design traffic is calculated as cumulative Million Standard Axles (MSA) over the design life (N):

$$N = \frac{365 \cdot A \cdot [(1+r)^n - 1]}{r} \cdot D \cdot F \cdot 10^{-6}$$

Where:

- N = Cumulative design traffic (MSA)
- A = Initial commercial traffic (CVPD) after construction $= P \cdot (1+r)^x$ (where P is count traffic and x is years between count and construction completion)
- r = Annual traffic growth rate decimal
- n = Design life (years)
- D = Lane Distribution Factor (LDF)
- Single-lane roads: LDF = 1.0
- Two-lane single carriageway: LDF = 0.50 (each direction)

- Four-lane single carriageway: LDF = 0.40
- Dual carriageway 2-lane: LDF = 0.75
- Dual carriageway 3-lane: LDF = 0.60
- Dual carriageway 4-lane: LDF = 0.45
- VF = Vehicle Damage Factor (VDF), converting commercial vehicles to equivalent standard axles.
- *Linear Case*: When growth rate $r = 0\%$, traffic accumulates linearly:

$$N = 365 * A * n * D * F * 10^{-6}$$

2.2 Subgrade Support (Resilient Modulus, M_r)

Subgrade resilient modulus (M_r) represents soil stiffness under repeated wheel loads. Per **IRC:37-2018 Annex E**:

- For soaked CBR (%) $\leq 5.0\%$:

$$M_r = 10 * CBR \text{ (in MPa)}$$

- For soaked CBR (%) $> 5.0\%$:

$$M_r = 17.6 * CBR^{0.64} \text{ (in MPa)}$$

- *Soil Cap Check*: Soil with soaked CBR $< 2.0\%$ must be replaced or stabilized with lime/cement.

2.3 Mechanistic Verification Strain Criteria

Pavements are evaluated at critical strain points under dual-wheel load (20,000 N per wheel at 0.8 MPa contact pressure).

Fatigue Life of Bituminous Layers (N_f)

Fatigue life represents the number of standard axle repetitions allowed before 20% cracking of the pavement surface. Sourced from **IRC:37-2018 Eq. 3.1 & 3.2**:

$$N_f = 0.0571 * \beta_f * \left(\frac{1}{\epsilon_t} \right)^{3.89} * \left(\frac{1}{E} \right)^{0.854}$$

Where:

- ϵ_t = Tensile strain at the bottom of the bituminous layer.
- E = Elastic modulus of the bituminous mix (MPa).
- β_f = Fatigue calibration factor (depends on binder type and air voids).

Rutting Life of Subgrade Soil (N_r)

Rutting life represents standard axle repetitions allowed before 20mm rut depth accumulates:

$$N_r = 1.41 \times 10^{-8} \times \beta_r \times \left(\frac{1}{\epsilon_v} \right)^{4.533}$$

Where:

- ϵ_v = Vertical compressive strain at the top of the subgrade.
- β_r = Rutting calibration factor.

3. CTB/CTS Stabilized Design Principles

Stabilized pavements introduce a Cement-Treated Base (CTB) and a Cement-Treated Sub-base (CTS) layer to increase load-bearing capability. Sourced from **IRC:37-2018 Section 9**:

3.1 Layer Configuration & Modulus

- **Bituminous Cover:** Sized to protect the stabilized slab from traffic wear and moisture infiltration.
- **CTB Modulus:** Sourced from laboratory flexural strength testing, typical design range: 3000 to 10000 MPa.
- **CTS Modulus:** Design range: 1000 to 4000 MPa.
- **GSB Layer:** Placed under the CTS to provide drainage and construction platforms.

3.2 Cracking Risks

Cement-treated materials are highly brittle and susceptible to:

1. **Shrinkage Cracking:** Occurs during cement hydration. Requires strict moisture control.
2. **Reflection Cracking:** Cracks in the CTB slab migrate upwards into the bituminous cover. Avoided by using SAMIs (Stress Absorbing Membrane Interlayers) or a granular sandwich layer.
3. **Tensile Fatigue:** Bottom tensile strain in the CTB slab must not exceed flexural capacity.

4. Marshall Mix Design (Volumetrics)

Marshall mix design determines the Optimum Binder Content (OBC) by checking the volumetric properties of compacted asphalt mixes (per **MORTH Section 500 / IRC:111**).

4.1 Volumetric Equations

- **Air Voids (V_a):**

$$V_a = \left(1 - \frac{G_{mb}}{G_{mm}} \right) \times 100$$

Where G_{mb} is bulk specific gravity and G_{mm} is maximum theoretical specific gravity of the compacted mix.

- **Voids in Mineral Aggregate (\$VMA\$):**

$$VMA = 100 - \frac{G_{mb}}{G_{mm}} \times P_s$$

Where $P_s = 100 - P_b$ is the aggregate percentage and G_{sb} is the combined aggregate specific gravity.

- **Voids Filled with Bitumen (\$VFB\$):**

$$VFB = \frac{VMA - V_a}{VMA} \times 100$$

4.2 OBC Spline Fitting

SamPave utilizes a cubic spline interpolation model to fit volumetric parameters against binder content (P_b). The OBC is determined as the binder percentage where V_a is exactly equal to 4.0% (target air voids), provided the mix satisfies MORTH limits for stability (> 9 kN), flow (2-4 mm), and VFB (65-75%).

5. Engineering Screening Compatibility Rules

The **Engineering Intelligence Checker** monitors layer layouts against compatibility guidelines:

- **Base-to-Subbase Modular Ratio:**

$$E_{base} / E_{sub-base} \leq 4.0$$

- **Stiff Layer Support:** Stiff stabilized base ($E \geq 3000$ MPa) must not sit directly on granular support (< 500 MPa), to prevent slab cracking.
- **Stiffness Inversions:** Lower layer elastic modulus must not exceed $1.3 \times E_{upper}$, preventing unbonded plate action.
- **Poisson's Ratio:** Restricted to realistic material bounds (granular = 0.35, stabilized = 0.25, bituminous = 0.35).